Notes on Linear Smoothers

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Chapter 1

Linear Smoothers

1.1 Least squares

A *linear smoother* is a method whose fitted values are a linear function of the response. Least squares is the oldest example (Gauss, 1809); a modern account is given by Hastie et al. (2009). Let X be the $N \times p$ input matrix. The fitted values are

$$\hat{y} = X(X^T X)^{-1} X^T y = Hy, \tag{1.1}$$

where H is called the *hat matrix*. The matrix in (1.1) is a projection.

Definition 1.1. The *degrees of freedom* of a linear smoother S is $\mathrm{trace}(S)$.

Lemma 1.2. *The hat matrix H in* (1.1) *is symmetric and idempotent.*

Proof. Direct computation gives $H^T = H$ and $H^2 = X(X^T X)^{-1} X^T X (X^T X)^{-1} X^T = H$. □

Theorem 1.3. *The degrees of freedom of least squares, in the sense of Definition 1.1, equals p.*

Proof. By Lemma 1.2 and the cyclic property of the trace,

$$\mathrm{trace}(H) = \mathrm{trace}\left((X^T X)^{-1} X^T X\right) = \mathrm{trace}(I_p) = p. \tag{1.2}$$

This completes the argument. □

1.2 Smoothing splines

Smoothing splines minimize a penalized criterion (Reinsch, 1967; Wahba, 1990). Their fit is again linear in y, so Theorem 1.3 generalizes through Definition 1.1.

$$\sum_{i=1}^{N}\{y_i - f(x_i)\}^2 + \lambda \int f''(t)^2\, dt. \tag{1.3}$$

Proposition 1.4. *The minimizer of* (1.3) *is a natural cubic spline with knots at the unique values of* x_i*.*

Proof. See Wahba (1990) for the classical argument; it uses (1.3) and an integration by parts. □

Figure 1.1 compares the two smoothers of Section 1.3.

Figure 1.1: A smoothing spline and a least squares line fitted to the same data.

1.3 Exercises

Exercise 1.1. Show that the trace in (1.2) equals the rank of X.

Exercise 1.2. Use Proposition 1.4 to compute the degrees of freedom of a smoothing spline.

Chapter 2

Kernel Methods

2.1 Local averaging

Kernel smoothers weight nearby observations (Nadaraya, 1964). The Nadaraya–Watson estimate is

$$\hat{f}(x_0) = \frac{\sum_{i=1}^{N} K_\lambda(x_0, x_i)\, y_i}{\sum_{i=1}^{N} K_\lambda(x_0, x_i)}, \tag{2.1}$$

where K_λ is a kernel with bandwidth λ. Like (1.1), it is a linear smoother.

Theorem 2.1. *The Nadaraya–Watson estimator* (2.1) *is a linear smoother whose degrees of freedom, as in Theorem 1.3, depend on* λ.

Proof. The weights in (2.1) do not depend on y, so by Definition 1.1 the claim follows from Lemma 1.2. □

Exercise 1.1 in Chapter 1 is useful here. Hastie et al. (2009) discuss bandwidth selection at length.

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